

A 28-VERTEX CIRCULANT WITH INTEGRAL FRACTIONAL BROADCAST NUMBER STRICTLY BETWEEN ITS MULTIPACKING AND BROADCAST NUMBERS

ABSTRACT. Problem 4 of the open-problems section of Teshima's multipacking survey asks whether some graph G has $\gamma_{b,f}(G)$ an integer with $\text{mp}(G) < \gamma_{b,f}(G) < \gamma_b(G)$. We exhibit such a graph. For the circulant $G = C_{28}(1, 4)$ on $\mathbb{Z}_{28}$,

$$\text{mp}(G) = 3 < \gamma_{b,f}(G) = 4 < \gamma_b(G) = 5,$$

and the middle term is the integer 4. Each of the three values is certified by hand: an explicit primal and an explicit dual of the broadcast linear program pin $\gamma_{b,f} = 4$ by weak duality, a ball-counting bound pins $\gamma_b = 5$, and the multipacking number is pinned between an exhibited 3-element multipacking and a mod-7 tiling obstruction. The dual is the instantiation at $r = 3$ of the vertex-transitive formula that is Theorem 10 of the survey itself; the only argument here that is not such an instantiation is the mod-7 obstruction pinning $\text{mp} = 3$. We make no claim that this is the first such witness: we did not determine the literature status of the problem, and did not read several of the most directly relevant items.

1. THE PROBLEM

Throughout, $G = (V, E)$ is a connected graph, $e(v)$ is the eccentricity of v , and $N_k[v] = \{u : d(u, v) \leq k\}$. A *broadcast* on G is a map $f : V \rightarrow \{0, 1, \dots, \text{diam}(G)\}$ with $f(v) \leq e(v)$ for all v ; it is *dominating* if every $u \in V$ lies in $N_{f(v)}[v]$ for some v with $f(v) \geq 1$; its *cost* is $\sigma(f) = \sum_v f(v)$, and $\gamma_b(G)$ is the least cost of a dominating broadcast. A set $M \subseteq V$ is a *multipacking* if

$$(1) \quad |M \cap N_s[v]| \leq s \quad \text{for every } v \in V \text{ and every } 1 \leq s \leq \text{diam}(G),$$

and $\text{mp}(G)$ is the largest size of a multipacking. Finally $\gamma_{b,f}(G)$ is the optimum of the broadcast linear program

$$(2) \quad \gamma_{b,f}(G) = \min \left\{ c \cdot x : Ax \geq \mathbf{1}, x \geq 0 \right\},$$

whose columns are the pairs (v, k) with $1 \leq k \leq e(v)$, whose rows are the vertices, with $A_{u,(v,k)} = 1$ iff $d(u, v) \leq k$, and with cost $c_{(v,k)} = k$. Linear programming duality gives $\text{mp} \leq \gamma_{b,f} \leq \gamma_b$. All of this is verbatim from the source below, whose notation we keep.

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Problem 4 (Teshima). Does there exist a graph G with integral $\gamma_{b,f}(G)$ such that $\text{mp}(G) < \gamma_{b,f}(G) < \gamma_b(G)$?

The statement is quoted verbatim, in the survey's own notation, from the fourth of the seven problems of the *Open Problems* subsection of the *Conclusions* of [2]; it carries no citation there.

2. THE WITNESS

Let $G = C_{28}(1, 4)$ be the circulant on $\mathbb{Z}_{28}$ in which i is joined to $i \pm 1$ and to $i \pm 4$. It is a connected, 4-regular, vertex-transitive graph with 28 vertices and 56 edges; in `graph6` it is the 64-byte line

`[hdHGcH@GC_H?H?C_@G?H??c?@G?@G??c??H??@G??C_??H_??HO??Cc??@K_??H`

Breadth-first search from 0 gives the levels

$$\{0\}, \quad \{1, 4, 24, 27\}, \quad \{2, 3, 5, 8, 20, 23, 25, 26\}, \quad \{6, 7, 9, 12, 16, 19, 21, 22\}, \\ \{10, 11, 13, 15, 17, 18\}, \quad \{14\},$$

so $e(v) = 5$ for every v , $\text{rad}(G) = \text{diam}(G) = 5$, and, writing $n_k = |N_k[v]|$,

$$(3) \quad (n_1, n_2, n_3, n_4, n_5) = (5, 13, 21, 27, 28).$$

Because $\text{rad} = \text{diam}$ here, the convention in (1) and the variant that ranges s only up to $\text{rad}(G)$, common elsewhere in the literature, agree on G .

Theorem 1. For $G = C_{28}(1, 4)$,

$$\text{mp}(G) = 3 < \gamma_{b,f}(G) = 4 < \gamma_b(G) = 5,$$

and $\gamma_{b,f}(G) = 4$ is an integer. So G has the property Problem 4 asks for.

The proof is four short certificates. Only (3), the level lists above and the single set

$$(4) \quad T := \{v : d(0, v) \geq 4\} = \{10, 11, 13, 14, 15, 17, 18\} = 14 + \{0, \pm 1, \pm 3, \pm 4\}$$

are used, with integer arithmetic throughout except for the two fixed rationals $1/21$ and $1/7$ in (A) and (B).

(A) $\gamma_{b,f}(G) \leq 4$. Put $x_{(v,3)} = \frac{1}{21}$ for all 28 vertices v and $x = 0$ on every other column. The column $(v, 3)$ is legal because $3 \leq e(v) = 5$. Each $u \in V$ satisfies $(Ax)_u = |N_3[u]|/21 = n_3/21 = 1$, so $Ax = \mathbf{1} \geq \mathbf{1}$, and the cost is $28 \cdot 3/21 = 4$.

(B) $\gamma_{b,f}(G) \geq 4$. Put $y_v = \frac{1}{7}$ for all v . For each column (v, k) , $1 \leq k \leq 5$,

$$\sum_{u \in N_k[v]} y_u = \frac{n_k}{7} = \frac{5}{7}, \frac{13}{7}, 3, \frac{27}{7}, 4 \quad \text{against} \quad c_{(v,k)} = k = 1, 2, 3, 4, 5,$$

so y is dual feasible, with equality exactly at $k = 3$; its objective is $\sum_v y_v = 28/7 = 4$. Weak duality alone now gives $\gamma_{b,f}(G) \geq 4$, and with (A), $\gamma_{b,f}(G) = 4$.

This dual is not ours. It is the instantiation at $r = 3$ of the vertex-transitive formula $\gamma_{b,f}(G) = n \cdot \min_{1 \leq r \leq \text{rad}(G)} r/n_r$, which is Theorem 10 of the source survey itself [2], where it is credited to R. C. Brewster and L. Duchesne [1]. Here $\min(1/5, 2/13, 1/7, 4/27, 5/28) = 1/7$, attained at $r = 3$, and $28 \cdot \frac{1}{7} = 4$.

(C) $\gamma_b(G) = 5$. Upper bound: $f(0) = 5$ and $f \equiv 0$ elsewhere is a legal broadcast, since $5 \leq e(0) = 5$ and $5 \leq \text{diam}(G)$, and it dominates because $N_5[0] = V$. So $\gamma_b(G) \leq 5$.

Lower bound, by counting only. A dominating broadcast of cost c assigns positive values $k_1, \dots, k_m$ with $\sum k_i = c$ and $1 \leq k_i \leq 5$, and covers at most $\sum_i n_{k_i}$ vertices. At $c = 4$ the multisets are

$$[4] \mapsto 27, \quad [3, 1] \mapsto 26, \quad [2, 2] \mapsto 26, \quad [2, 1, 1] \mapsto 23, \quad [1, 1, 1, 1] \mapsto 20,$$

with maximum $27 < 28$; and the same recursion gives maxima 5, 13, 21 at $c = 1, 2, 3$, all below 28. So no broadcast of cost at most 4 dominates, and $\gamma_b(G) = 5$.

(D) $\text{mp}(G) = 3$. *Lower bound.* $M = \{0, 6, 15\}$ has pairwise distances 3, 3, 4. Over all 28 vertices v , $\max_v |M \cap N_1[v]| = 1$ and $\max_v |M \cap N_2[v]| = 2$; concretely $N_2[0] \cap N_2[6] = \{1, 2, 3, 4, 5, 8, 26\}$, which is disjoint from $N_2[15] = \{7, 10, 11, \dots, 20, 23\}$, so no vertex has all three of M within distance 2. For $s \geq 3$ the condition (1) is vacuous since $|M| = 3$. Hence M is a multipacking and $\text{mp}(G) \geq 3$.

Upper bound. Suppose M is a multipacking with $|M| = 4$. Summing the $s = 3$ case of (1) over all v and exchanging the order of summation,

$$\sum_{v \in V} |M \cap N_3[v]| = \sum_{u \in M} |N_3[u]| = 4 \cdot 21 = 84 \leq 28 \cdot 3 = 84.$$

Equality is forced, so $|M \cap N_3[v]| = 3$ for *every* v : each vertex v has exactly one $u \in M$ with $d(u, v) \geq 4$. Writing $F_u = \{v : d(u, v) \geq 4\}$, the four sets F_u ($u \in M$) therefore partition $\mathbb{Z}_{28}$, and $4 \cdot 7 = 28$ is consistent. By vertex-transitivity of the circulant, $F_u = u + T$ with T as in (4).

Reduce modulo 7. Let $t_j = |\{\tau \in T : \tau \equiv j \pmod{7}\}|$, so

$$(t_0, \dots, t_6) = (1, 1, 0, 2, 2, 0, 1),$$

and let $m_j = |\{u \in M : u \equiv j \pmod{7}\}|$, so $\sum_j m_j = 4$. Each residue class of $\mathbb{Z}_{28}$ modulo 7 contains exactly 4 elements, and the tiling says every $x \in \mathbb{Z}_{28}$ has exactly one representation $x = u + \tau$ with $u \in M$, $\tau \in T$; counting such pairs inside a fixed class gives

$$(5) \quad (m * t)(j) := \sum_{a \in \mathbb{Z}_7} m_a t_{j-a} = 4 \quad \text{for every } j \in \mathbb{Z}_7.$$

Now $t(x) = 1 + x + 2x^3 + 2x^4 + x^6$ is monic of degree 6, and so is $\Phi_7(x) = 1 + x + \dots + x^6$, which is irreducible over $\mathbb{Q}$. If t vanished at a primitive 7th root of unity ζ then $\Phi_7 \mid t$, and equality of degrees with both monic would force $t = \Phi_7$ — false, since t has coefficient 0 at x^2 . So $t(\zeta) \neq 0$ for

every nontrivial character of $\mathbb{Z}_7$. The right-hand side of (5) is the constant vector 4, whose Fourier transform is supported on the trivial character alone; hence $\hat{m}$ vanishes at every nontrivial character, m is constant, and $7m_0 = 4$ — impossible over the integers. Therefore no 4-element multipacking exists and $\text{mp}(G) \leq 3$.

Proof of Theorem 1. Combine (A)–(D): $\text{mp}(G) = 3$, $\gamma_{b,f}(G) = 4$, $\gamma_b(G) = 5$. The chain is strict and the middle term is an integer, which is the property Problem 4 asks for. G is connected, as the problem requires. $\square$

3. SCOPE

Status and priority. We make no priority claim, and we do not establish that Problem 4 was still unanswered: several of the most directly relevant items — among them the manuscript [1] to which the survey credits the formula that certificate (B) instantiates — we did not read. What is offered here is an explicit witness with hand certificates; whether it is the first, an independent one, or a rediscovery we do not know.

Minimality. We do not claim that G has least order among graphs satisfying Problem 4, and nothing here bounds that order from below.

Verification. The program `verify.py` accompanying this note re-derives, in exact integer and rational arithmetic and from the `graph6` string and the definitions printed above, every quantity claimed in Theorem 1: the ball profile, the primal and dual feasibility of (A) and (B) at all 140 constraints, the counting bound of (C), and both halves of (D) — the latter three independent ways, by the polynomial argument above, by an exhaustive scan of all 210 residue distributions, and by an exhaustive scan of all 2,925 four-element subsets containing 0 (legitimate because $\mathbb{Z}_{28}$ acts transitively). It covers this graph and nothing else, and states in its own output what it does not cover.

REFERENCES

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